

PHY202: Problem Set 5

Students with odd (even) Reg. numbers should submit the solution to problem number 3 (5) in the beginning of the class on Wednesday, Oct. 29th.

1. The three Pauli matrices, which are very useful in quantum mechanics, are given by,

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Prove the following relations:

- (i) $\sigma_1^2 = \sigma_2^2 = \sigma_3^2 = \text{Identity Matrix}$
 - (ii) $\sigma_i \sigma_j + \sigma_j \sigma_i = 0$ for all pairs $i \neq j$
 - (iii) $\sigma_i \sigma_j - \sigma_j \sigma_i = 2\sigma_k$, where i, j, k are in cyclic order.
2. Check the linear dependence/independence of the following functions:
- (i) x, x^2, x^3
 - (ii) x, e^x, xe^x
 - (iii) $\sin x, \sin 2x$
3. Consider a general triangle with a, b, c as the lengths of the sides. Let α, β, γ be the angles opposite to a, b, c , respectively.
- (i) Express the length of each side in terms of the projections of other two sides onto it.
 - (ii) Write the resulting 3 equations in the matrix form $Ax = k$, with $x^T = (\cos \alpha \ \cos \beta \ \cos \gamma)$.
 - (iii) Solve the matrix equation and find the expression for $\cos \alpha$ in terms of a, b, c .
4. Show that: (i) AA^T is a symmetric matrix. (ii) $(AB)^\dagger = B^\dagger A^\dagger$
5. Consider the matrix, $M = \begin{pmatrix} 1 & 0 \\ 3 & -2 \end{pmatrix}$. (i) Find the eigenvalues and eigenvectors of M .
(ii) Is the matrix diagonalizable? (iii) Find the matrix, $\sin(\pi M/2)$.